Remainders modulo 4 for the series with $A\big(x\,A(x)\big) = \theta_3(x)$

A proof of a conjecture in OEIS A378581

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Abstract

Let $\theta(x) = 1 + 2x + 2x^4 + 2x^9 + 2x^{16} + \cdots$ be the theta series, whose coefficient of x^n is 2 when n is a positive perfect square and 0 otherwise. Entry A378581 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna, lists the coefficients $a(n)$ of the power series $A(x) = 1+2x-4x^2+24x^3-206x^4+\cdots$ determined by $A\big(x\,A(x)\big) = \theta(x)$. The entry conjectures that for $n > 0$ the coefficient $a(n)$ leaves remainder 2 on division by 4 when n is a perfect square, and is divisible by 4 otherwise. We prove this conjecture. In fact, modulo 4 the whole series $A(x)$ agrees with $\theta(x)$, and the same is true for the companion equation $A\big(x/A(x)\big) = \theta(x)$ of entry A378580. The proof rests on one observation: modulo 4, twice a series does not change when its argument is changed by a multiple of 2. The results were checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [4] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record *conjectures*: patterns that the contributor observed in the data but could not prove.

This paper is about entry A378581 [1], contributed by Paul D. Hanna in January 2025. It starts from the *theta series*

$$\theta(x) = 1 + 2\sum_{j\ge 1} x^{j^2} = 1 + 2x + 2x^4 + 2x^9 + 2x^{16} + 2x^{25} + \cdots .$$

Another way to write it is $\theta(x) = \sum_k x^{k^2}$, where k runs over all integers, positive, negative and zero: each positive square j^2 appears twice, once from $k = j$ and once from $k = -j$. So

the coefficient of x^n in $\theta(x)$ is the number of integers whose square is n. This series is one of the classical theta functions studied by Jacobi, and the OEIS writes it as `theta_3`.

The entry defines a power series $A(x) = a(0) + a(1)x + a(2)x^2 + \cdots$, with $a(0) = 1$, by the equation

$$A(x\,A(x)) = \theta(x). \tag{1}$$

Here $x\,A(x)$ is a series with constant term 0, and the left side of (1) substitutes it for x in A. The entry reads:

> `G.f. A(x) satisfies: A(x*A(x)) = theta_3(x) = 1 + 2*Sum_{n>=1} x^(n^2).`
>
> `Conjecture: for n > 0, a(n) == 2 (mod 4) iff n is square, else a(n) is divisible by 4 if n is nonsquare.`
>
> — OEIS A378581, Paul D. Hanna, January 2025

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence.

Equation (1) determines the coefficients one at a time. Write $A = 1 + a(1)x + a(2)x^2 + \cdots$. Then $x\,A = x + a(1)x^2 + \cdots$, and

$$A(xA) = 1 + a(1)\big(x + a(1)x^2\big) + a(2)x^2 + \cdots = 1 + a(1)\,x + \big(a(2) + a(1)^2\big)x^2 + \cdots .$$

Comparing with $\theta(x) = 1 + 2x + 0x^2 + \cdots$ gives $a(1) = 2$ and $a(2) = -4$. In the same way, the coefficient of x^n in (1) determines $a(n)$ once the earlier coefficients are known (Lemma 4.1). So there is exactly one solution, and all its coefficients are integers. Its first terms are 1, 2, -4, 24, -206, 2188, -26576, 356256, -5160752, 79686506, … (starting from $n = 0$). The signs of the terms alternate. As usual, the remainder of a negative number on division by 4 is taken in $\{0, 1, 2, 3\}$; for example $-206 = 4 \cdot (-52) + 2$ leaves remainder 2.

1.2 The question

The conjecture says that for $n > 0$, the coefficient $a(n)$ leaves remainder 2 on division by 4 when n is a perfect square ($n = 1, 4, 9, 16, \ldots$), and remainder 0 when n is not a square. (Section 2 recalls the notation $a \equiv b \pmod 4$ for "a and b leave the same remainder on division by 4".) The table shows the remainders of the first terms. The indices in bold are the positive squares.

n	0	**1**	2	3	**4**	5	6	7	8	**9**	10	11	12	13	14	15
$a(n) \bmod 4$	1	**2**	0	0	**2**	0	0	0	0	**2**	0	0	0	0	0	0
n	**16**	17	18	19	20	21	22	23	24	**25**	26	27	28	29	30	31
$a(n) \bmod 4$	**2**	0	0	0	0	0	0	0	0	**2**	0	0	0	0	0	0
n	32	33	34	35	**36**	37	38	39	40	41	42	43	44	45	46	47
$a(n) \bmod 4$	0	0	0	0	**2**	0	0	0	0	0	0	0	0	0	0	0

Every term in the table follows the conjectured pattern.

1.3 What this paper proves

Main result (Theorem 3.1). *Modulo* 4*, the series* $A(x)$ *is equal to* $\theta(x)$*: every coefficient* $a(n)$ *leaves the same remainder on division by* 4 *as the corresponding coefficient of* $\theta(x)$*.*

Since the coefficients of $\theta(x)$ are 1 at $n = 0$, 2 at the positive squares and 0 elsewhere, this is exactly Hanna's conjecture (Corollary 3.2). The theorem also covers the companion entry A378580 [2], defined by $A\big(x/A(x)\big) = \theta(x)$, which has the same conjecture. The two papers on these entries share all of their mathematics.

1.4 What was known

The entry records several formulas due to Hanna, for example $A(x) = \theta\big(B(x)\big)$, where B is the compositional inverse of $x\,A(x)$. These formulas do not seem to give the remainders directly. We are not aware of an earlier proof of the conjecture.

1.5 Why it is interesting

Theta series are among the most important objects in number theory. For instance, the coefficient of x^n in $\theta(x)^k$ is the number of ways to write n as a sum of k squares of integers, and classical theorems of Jacobi give exact formulas for these numbers when $k = 2$ and $k = 4$ [3]. Hanna's entries explore what happens when such classical series are combined with substitution. The result here says that, modulo 4, the substitution in (1) is invisible: the solution is θ itself.

The reason is simple and general (Section 6): the argument works for every target series of the form $1 + 2S$, and the same argument handles both entries at once. The key tool, Lemma 2.1, is a simple way of passing from information modulo 2 to information modulo 4; it is used repeatedly in this collection of papers.

1.6 How to read this paper

Section 2 explains the few notions we need: power series, substitution, and remainders. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result, a generalization, and a further congruence between the two companion entries.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed. Standard references for formal power series are [6, 5].

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0v_n + u_1v_{n-1} + \cdots + u_nv_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots)=1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D\cdot\frac{1}{D}=1$ using only addition, subtraction and multiplication. Every fraction in this paper has a denominator of this kind, so it denotes a power series.

We write $U = O(x^d)$ when all coefficients of U below degree d are zero. So $U - V = O(x^d)$ means that U and V agree up to degree $d-1$. A series of the form $x + u_2x^2 + u_3x^3 + \cdots$, with constant term 0 and coefficient of x equal to 1, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1V(x) + u_2V(x)^2 + u_3V(x)^3 + \cdots.$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U\cdot U$, $U^3 = U\cdot U\cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a-b$ is divisible by m, that is, when a and b leave the same remainder after division by m. For example, $17 \equiv 5 \pmod 4$, since both leave remainder 1, and $-3 \equiv 1 \pmod 4$. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$. Saying that a is odd is the same as saying $a \equiv 1 \pmod 2$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod m$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Several lemmas below hold over every *commutative ring*, that is, every set with an addition, subtraction and multiplication satisfying the usual rules of arithmetic (for example

$\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$). No division is used; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.4 The method

Almost every proof in this paper follows the same three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/4\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo 4 gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A good guess.* Find a simple series that satisfies the reduced equation and check that it does. By uniqueness, this simple series *is* the reduction of the sequence, and its coefficients can simply be read off.

The creative part is step 3: finding a simple series that works. Step 1 sometimes needs care, when the equation seems to require a division.

2.5 Doubling modulo 4

Lemma 2.1. *Let U, V be series with coefficients in $\mathbb{Z}/4\mathbb{Z}$. Then $2U = 2V$ if and only if U and V agree modulo 2. Consequently, if P and Q have constant term 0 and $P \equiv Q \pmod 2$, then $2H(P) = 2H(Q)$ in $\mathbb{Z}/4\mathbb{Z}$ for every series H.*

Proof. For residues u, v modulo 4, $2u = 2v$ means that 4 divides $2(u - v)$, that is, 2 divides $u - v$. Apply this to every coefficient. For the second claim, $H(P)$ and $H(Q)$ agree modulo 2 because reduction is compatible with substitution. □

In other words, *twice a series, modulo 4, depends only on the series modulo 2.* This lets us use information modulo 2 to get information modulo 4, without ever dividing by 2 in $\mathbb{Z}/4\mathbb{Z}$, which is impossible.

2.6 A one-term Taylor formula

Lemma 2.2. *Let ε be an element of a commutative ring with $\varepsilon^2 = 0$. If u and v are series with constant term 0, then for every series S*

$$S(u + \varepsilon v) = S(u) + \varepsilon\, v\, S'(u),$$

where $S' = \sum_{n\geq 1} n\, s_n x^{n-1}$ is the formal derivative of S.

Proof. By the binomial theorem, $(u+\varepsilon v)^j = u^j + j\, \varepsilon v\, u^{j-1}$, because all terms with ε^2 or higher powers vanish. Multiply by s_j and add up over j. □

An example of such an ε: in $\mathbb{Z}/8\mathbb{Z}$, the element 4 satisfies $4^2 = 16 = 0$.

3 Results

Theorem 3.1. *Each of the two equations*

$$A(x/A(x)) = \theta(x) \qquad \text{and} \qquad A(x\,A(x)) = \theta(x)$$

has exactly one solution A with integer coefficients and $A(0) = 1$. These solutions are the generating functions of A378580 and A378581, respectively. Both satisfy

$$A(x) \equiv \theta(x) = 1 + 2\sum_{j\geq 1} x^{j^2} \pmod 4.$$

Corollary 3.2 (Hanna's conjecture)**.** *For the sequence $a(n)$ of OEIS A378581 and every $n > 0$: if n is a perfect square, then $a(n) \equiv 2 \pmod 4$; otherwise $a(n) \equiv 0 \pmod 4$, that is, 4 divides $a(n)$.*

Proof. Compare the coefficients of x^n on both sides of the congruence in Theorem 3.1. For $n > 0$ the coefficient of x^n in $\theta(x)$ is 2 if n is a perfect square and 0 otherwise. □

4 Proofs

The idea in brief. Work modulo 4 and write $\theta = 1 + 2S$, with $S = x + x^4 + x^9 + \cdots$. Then $\theta^2 = 1 + 4S + 4S^2 = 1$, so θ is its own multiplicative inverse, and $x/\theta = x\theta$. Both equations therefore ask whether $\theta(x\theta) = \theta$. Now $x\theta = x + 2xS$ differs from x by a multiple of 2. Substituting it into $\theta = 1 + 2S$ changes S by a multiple of 2, so it changes $2S$ by a multiple of 4, that is, not at all modulo 4. So θ solves both equations modulo 4, and by uniqueness it is the reduction of both solutions.

Lemma 4.1 (Existence and uniqueness)**.** *Let R be a commutative ring and let T be a series with coefficients in R and constant term 1. Each of the equations*

$$U(x\,U(x)) = T(x) \qquad \text{and} \qquad U(x/U(x)) = T(x)$$

has exactly one solution U with coefficients in R and constant term 1.

Proof. Write $U = 1 + u_1x + u_2x^2 + \cdots$ and $T = 1 + t_1x + t_2x^2 + \cdots$.

The product form. Since $U(y) = \sum_k u_k y^k$, we have $U(xU) = \sum_{k\geq 0} u_k\, x^k U^k$, and the term with index k has no terms below degree k. So for $n \geq 1$ the coefficient of x^n in $U(xU)$ is

$$u_n \cdot [x^0]\, U^n + \sum_{k=1}^{n-1} u_k\, [x^{n-k}]\, U^k = u_n + \big(\text{an expression in } u_1, \ldots, u_{n-1}\big),$$

because $[x^0]\, U^n = 1$, and $[x^{n-k}]\, U^k$ involves only $u_1, \ldots, u_{n-k}$, where $n - k \leq n - 1$. (The term $k = 0$ is the constant 1.) Setting this coefficient equal to t_n determines u_n from $u_1, \ldots, u_{n-1}$, using only addition, subtraction and multiplication. So the coefficients are determined one at a time: there is exactly one solution.

The quotient form. Put $W = 1/U$, the multiplicative inverse of U, so that $x/U = xW$. Its coefficients are determined by $UW = 1$, that is, $w_j = -(u_1w_{j-1} + u_2w_{j-2} + \cdots + u_jw_0)$, so w_j depends only on $u_1, \ldots, u_j$. Now $U(xW) = \sum_k u_k\, x^k W^k$, and exactly as before, the coefficient of x^n is u_n plus an expression in $w_1, \ldots, w_{n-1}$ and $u_1, \ldots, u_{n-1}$, hence in $u_1, \ldots, u_{n-1}$ alone. So again the coefficients are determined one at a time. □

No division was used in this proof. So, as explained in Section 2, reducing the integer solution modulo 4 gives the solution over $\mathbb{Z}/4\mathbb{Z}$, and to find that reduction it is enough to find *some* solution over $\mathbb{Z}/4\mathbb{Z}$.

Proof of Theorem 3.1. The first two claims are Lemma 4.1 with $R = \mathbb{Z}$ and $T = \theta$. For the congruence, work over $\mathbb{Z}/4\mathbb{Z}$ and write $\theta = 1 + 2S$ with $S = \sum_{j \geq 1} x^{j^2}$.

Step 1. $\theta^2 = 1 + 4S + 4S^2 = 1$, because $4 = 0$ in $\mathbb{Z}/4\mathbb{Z}$. So $1/\theta = \theta$, and therefore $x/\theta = x\theta$.

Step 2. The series $x\theta = x + 2xS$ agrees with x modulo 2, and both have constant term 0. By Lemma 2.1, $2S(x\theta) = 2S(x)$. Hence

$$\theta(x\theta) = 1 + 2S(x\theta) = 1 + 2S(x) = \theta(x).$$

Step 3. By Steps 1 and 2, $\theta(x/\theta) = \theta(x\theta) = \theta$. So $U = \theta$ solves both equations of Lemma 4.1 over $\mathbb{Z}/4\mathbb{Z}$, with $T = \theta$. By uniqueness, the reduction modulo 4 of each integer solution is θ. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed A from the defining equation (1), with exact integer arithmetic, through degree 199. This reproduces all 22 terms printed in the entry, and for every $0 < n < 200$ the remainder of $a(n)$ modulo 4 is 2 when n is a perfect square and 0 otherwise. The companion sequence A378580 was checked in the same way. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [9]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [7] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [8], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A378581.lean` of the repository, in the namespace `OEIS.A378581`. There θ is `theta`, and x/A is written `X * invOfUnit A 1`, the product of x with the multiplicative inverse of A. The general constructions, for an arbitrary series T with constant term 1 over any commutative ring, are in `lean/Families/ThetaSubstitution.lean`. The Lean proof of Lemma 4.1 takes a slightly different route from the one above: it treats the product form through the equivalent equation $G(G(x)) = G(x)\,T(x)$ for $G = xU$, and reduces the quotient form to the product form using $U = 1/V$. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
$\theta(0) = 1$, and for $n > 0$ the coefficient of x^n in θ is 2 or 0 as n is or is not a square	`theta_constant_term`, `theta_coefficient_formula`
$A(0) = 1$ and $A(xA) = \theta$, and Lemma 4.1 for this equation	`plus_constant`, `plus_source`, `plus_exists_unique`, `plus_unique`
Theorem 3.1 ($A \equiv \theta \pmod 4$) and Corollary 3.2	`plus_series_mod_four`, `plus_congruence`, `plus_square_iff`, `plus_four_dvd_iff`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Theorem 3.1 describes the remainder modulo 4 of every coefficient, so it settles the conjecture completely, for this entry and for its companion A378580. It says nothing about the size of the coefficients.

A generalization. The proof used only that $\theta = 1 + 2S$ for some series S with integer coefficients and $S(0) = 0$. So the same three steps show: *for every such target series $T = 1+2S$, the solutions of $U(xU) = T$ and of $U(x/U) = T$ with constant term 1 are both congruent to T modulo 4.* As a check, we confirmed this by computer for 20 randomly chosen targets T, up to degree 59. This general form is not part of the Lean formalization.

The two entries agree modulo 8. Neither solution agrees with θ modulo 8: for both entries $a(2) = \pm 4$, while the coefficient of x^2 in θ is 0. But the two sequences A378580 and A378581 agree with *each other* modulo 8. More generally, for every series T with integer coefficients and $T \equiv 1 \pmod 2$, the solutions of $U(x/U) = T$ and $U(xU) = T$ are congruent modulo 8. Here is the argument, which is not part of the Lean formalization. Work over $\mathbb{Z}/8\mathbb{Z}$, and let U solve $U(xU) = T$. Modulo 2 we have $T = 1$, and then $U = 1$ is the solution, so $U = 1 + 2W$ for some series W. Then $U^2 = 1 + 4(W + W^2)$, and since $(1 + 4X)^2 = 1 + 8X + 16X^2 = 1$ in $\mathbb{Z}/8\mathbb{Z}$, we get $1/U^2 = 1 + 4(W + W^2)$ and

$$\frac{x}{U} = x\,U \cdot \frac{1}{U^2} = x\,U + 4E, \qquad E = x\,U\,(W + W^2).$$

By Lemma 2.2 with $\varepsilon = 4$ (note $4^2 = 16 = 0$ in $\mathbb{Z}/8\mathbb{Z}$), $U(xU + 4E) = U(xU) + 4E\,U'(xU)$. The derivative $U' = 2W'$ is a multiple of 2, so the last term is a multiple of 8, which is 0. Hence $U(x/U) = U(xU) = T$: the same U solves the quotient equation, and by Lemma 4.1 the two solutions agree modulo 8. The two sequences do not agree modulo 16: already $a(2) = 4$ for A378580 and $a(2) = -4$ for A378581, which differ by 8. A computer check confirms the agreement modulo 8 for every $n < 160$.

Context. Section 2 explained the general method: reduce modulo a small number, find a simple solution of the reduced equation, and conclude by uniqueness. Here the simple solution is the target itself. The same doubling argument (Lemma 2.1) appears in other papers of this collection, for example for the theta-type sums of entries A380060 and A380680.

About this paper

How the work was done. This paper is one of 26 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [9]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [9], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence, its companion A378580, and the conjecture are due to Paul D. Hanna [1, 2].

References

[1] P. D. Hanna, entry A378581 (G.f. $A(x)$ satisfies $A\big(x\,A(x)\big) = \theta(x)$), *The On-Line Encyclopedia of Integer Sequences*, January 2025. `https://oeis.org/A378581`. Text as of October 2026.

[2] P. D. Hanna, entry A378580 (G.f. $A(x)$ satisfies $A(x/A(x)) = \theta_3(x)$), *The On-Line Encyclopedia of Integer Sequences*, January 2025. `https://oeis.org/A378580`. Text as of October 2026.

[3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, sixth edition, Oxford University Press, 2008. (Chapter 20 treats sums of two and four squares.)

[4] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[5] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[6] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

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